# MOSAIC: Data-Certified Stochastic Release under Structured Deployment Shift

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Code and artifacts: https://github.com/RudraChopra/vera-edit-or-abstain/tree/mosaic-aaai2027-v1

**Abstract**

Probe failure on one validation split does not guarantee that an edited representation hides a source attribute from every downstream rule after deployment shift. We present MOSAIC, a software-only method that jointly selects a task-specific stochastic release channel and decoder, or abstains. One simultaneous confidence region over pre-release token laws covers the entire same-table continuum of channels. A new bridge program uses labeled target data to certify the largest retained mass explained by a common Markov transform; arbitrary source-specific contamination is charged to exact channel capacity. We prove exact finite-sample envelopes for the Bayes source attacker within every task label and worst-stratum task error, global solvability for finite alphabets, and impossibility under unrestricted differential shift or a missing target source. In 1,000 hash-locked trials at a safety boundary, naive continuum selection deploys contract-violating releases on 42.1% of tables; MOSAIC records none. On 1,000 paired low-data tables, using exact shift geometry raises safe deployment from 0.3% to 53.0%, with zero observed false acceptances. Every decision and population risk is retained for replay.

## 1 Introduction

Representation editing is commonly evaluated by fitting a probe to a frozen embedding (Ravfogel et al. 2020, 2022; Belrose et al. 2023; Jourdan et al. 2023; Avitan, Goldberg, and Elazar 2026), although probe design can change the apparent conclusion (Pimentel et al. 2020). This leaves two deployment gaps: a different downstream rule may recover the attribute, and an edit selected on a noisy validation table may fail after population shift. Both matter when the released representation is a public interface.

We study a task-specific finite-token interface. A tokenizer fixed without the certification fold maps an internal representation to $C \in [K]$; a source-blind stochastic channel $M \in \mathrm{Stoch}(K, L)$ releases the only public token $Z$. Within each task label, the contract bounds the balanced accuracy of the Bayes-optimal source attacker, not a chosen probe. It does not claim to hide source information carried only by label prevalence. Pre-release shift is a common Markov transform plus bounded source-specific contamination, permitting large source-blind drift while charging the part that can create new source differences.



MOSAIC, Minimax-Optimized Source-Agnostic Invariant Channels, builds an exact attacker envelope from simultaneous multinomial confidence regions. Because the event precedes $M$, it covers every stochastic channel, including one optimized on the same table. A bridge LP uses labeled target data to learn a common transform and the largest certifiable retained mass; exact transform-and-contamination envelopes then certify source distinguishability and a jointly selected decoder. Finite-alphabet optimization is global, and no feasible release returns ABSTAIN.

Randomized channels, confidence regions, contraction, and abstention are not individually new. The contribution is their shift-aware composition: an exact same-table universal-attacker certificate, a data-certified target bridge, matching robust utility, sharp impossibility boundaries, and a globally solved release-or-abstain rule. Hash-locked experiments compare plug-in and held-out selection, LTT, deterministic release, shift-unaware certification, official erasers, and an exact population oracle.

## 2 Related Work

**Erasure and universal representations.** Concept-erasure methods remove source-associated directions, covariance, or components (Ravfogel et al. 2020, 2022; Belrose et al. 2023; Jourdan et al. 2023; Avitan, Goldberg, and Elazar 2026); interventions and leakage audits explain why a weak probe is not a universal certificate (Elazar et al. 2021; Chowdhury et al. 2025). FARE certifies every downstream classifier for restricted finite encoders, while Fair Normalizing Flows bounds strong adversaries through tractable densities (Jovanovic et al. 2023; Balunović, Ruoss, and Vechev 2022). MOSAIC instead places its confidence event on pre-release token laws, uniformly covering a stochastic post-channel optimized on that table and transferred through an explicit shift class.

**Randomized mappings and risk control.** Probabilistic preprocessing and fair representation already show the value of stochastic channels (Calmon et al. 2017; Makhdoumi et al. 2014; Agarwal and Deshpande 2024). Risk-controlling prediction, Learn-Then-Test (LTT), conformal risk control, and Pareto testing certify registered decisions and may abstain (Bates et al. 2021; Angelopoulos et al. 2025, 2024; Laufer-Goldshtein et al. 2023). Robust variants address specified

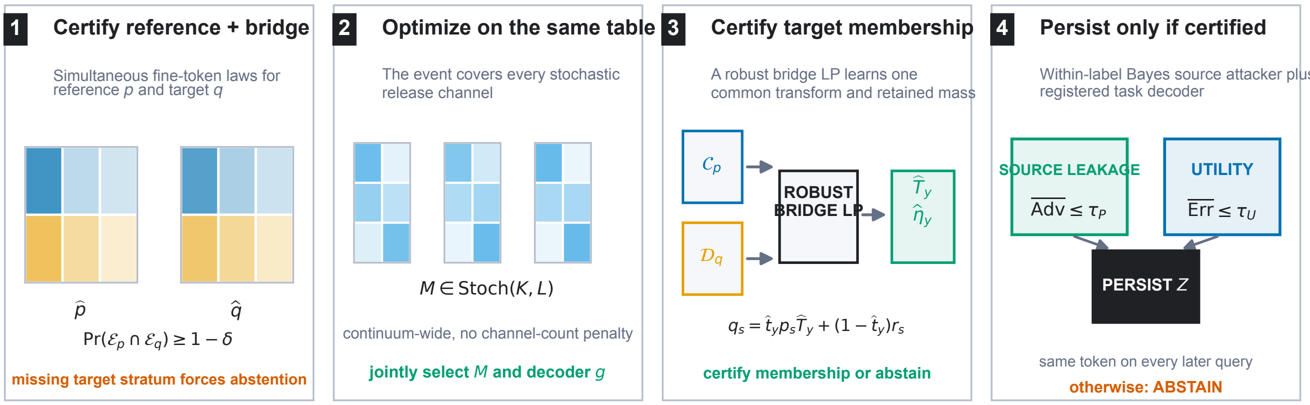


Figure 1: MOSAIC jointly certifies reference and labeled target-bridge token laws, optimizes a continuum of source-blind stochastic channels on that event, and uses a robust bridge LP to certify target membership. The runtime samples one persistent token only when within-label source distinguishability and registered task utility pass; otherwise it abstains. The primary contract is within-label source distinguishability.

covariate or uncertainty shifts (Tibshirani et al. 2019; Cauchois et al. 2024). These are the closest decision-layer precedents. MOSAIC instead covers an uncountable channel family through one confidence region for its sufficient table, without testing channels separately.

**Shift, comparison of experiments, and abstention.** Dobrushin coefficients quantify channel contraction (Diaz, Sankar, and Kairouz 2018); fairness and domain-adaptation impossibility results show why unrestricted shift precludes universal guarantees (Lechner et al. 2021; Agarwal et al. 2025; David et al. 2010). WILDS and subgroup-robustness benchmarks document the practical problem (Koh et al. 2021; Sagawa et al. 2020; Zhou et al. 2021; Sohoni et al. 2020). Bounded-shift fairness transfer and distributional fairness certification give complementary predictor-level robust bounds (Chen et al. 2022; Kang et al. 2022). Common source-blind transformation is Blackwell garbling, while Le Cam deficiency handles approximate experiment comparison (Blackwell 1953; Le Cam 1964; Torgersen 1991); arbitrary residuals follow contamination neighborhoods (Huber 1964). Testable learning asks when target data can validate a shift assumption (Klivans, Stavropoulos, and Vasilyan 2024; Chandrasekaran et al. 2025). MOSAIC contributes a finite-confidence garbling-and-contamination bridge that composes with the same-table optimizer. Its contract-driven abstention is distinct from predictive-confidence rejection (El-Yaniv and Wiener 2010; Geifman and El-Yaniv 2017).

## 3 Problem Setup

Let $S \in [G]$ denote source, $Y \in [J]$ task label, and $C \in [K]$ the fine token. For a fixed label, write $p_{s,y}(c) = \Pr(C = c \mid S = s, Y = y)$. The source-blind release channel $M$ produces $Z \in [L]$. The balanced accuracy of the strongest downstream source attacker within label $y$ is

$$A_y(P, M) = \frac{1}{G} \sum_{z=1}^{L} \max_{s \in [G]} (p_{s,y} M)(z), \tag{1}$$

with normalized advantage $\mathrm{Adv}_G(A) = (GA-1)/(G-1)$. This is chance-normalized and ranges from zero to one. The registered source contract is $\max_y \mathrm{Adv}_G(A_y) \leq \tau_P$.

This target must not be read as unconditional source privacy. To certify an attacker on the natural mixture, define the joint fine symbol $X = (Y, C)$, estimate $P(X \mid S)$ without balancing labels, and lift the channel as $\widetilde{M}((y,c), z) = M(c, z)$. The same envelope below then certifies source inference from $Z$; using output $(Y, Z)$ certifies the stronger attacker also given the true label. Our experiments use the registered within-label target, so they do not estimate this prevalence channel.

For each supported source–label stratum, the certification table supplies an empirical law $\widehat{p}_{s,y}$ and radius $\epsilon_{s,y}$. The simultaneous confidence set is $\mathcal{C}_{s,y} = \{p \in \Delta_K : \|p - \widehat{p}_{s,y}\|_1 \leq \epsilon_{s,y}\}$. Under fixed-size i.i.d. or exchangeable sampling within each registered stratum, the simultaneous event

$$\mathcal{E} = \bigcap_{s,y} \{\|\widehat{p}_{s,y} - p_{s,y}\|_1 \leq \epsilon_{s,y}\} \tag{2}$$

has probability at least $1-\delta$ after a fixed allocation across strata. We use the Weissman multinomial radius (Weissman et al. 2003) in the confirmation, although any valid simultaneous region can replace it. The tokenizer and fine alphabet must be fixed without these observations.

The external law for label $y$ belongs to the structured class

$$q_s = t_y p_s T_y + (1-t_y) r_s, \qquad t_y \geq 1 - \eta_y, \tag{3}$$

where $T_y$ is a source-independent fine-token channel from a registered convex uncertainty set and each residual $r_s$ is arbitrary. Both components pass through the same release channel $M$. This placement is crucial: a constant-row software channel cannot leak information that appears before it.

| Method | Finite $Z$ | Bayes attacker | Same-table $M$ | Target shift | Shift evidence |
|---|---|---|---|---|---|
| FARE (Jovanovic et al. 2023) | yes | yes | no | no | no |
| Randomized fair reps. (Agarwal and Deshpande 2024) | yes | yes | population | no | no |
| LTT / risk control (Angelopoulos et al. 2025; Bates et al. 2021) | general | candidate | finite family | optional | model-specific |
| Blackwell / deficiency (Blackwell 1953; Le Cam 1964) | experiment | all decisions | no | population | assumed |
| Robust validation (Cauchois et al. 2024) | no | no | no | declared ball | estimated |
| Testable shift (Klivans, Stavropoulos, and Vasilyan 2024) | no | no | hypothesis | yes | tested |
| MOSAIC | yes | exact | yes | pre-release | finite-sample |

Table 1: Component-level scope of the closest frameworks. "Same-table continuum" means that one confidence event remains valid after optimizing an uncountable stochastic-channel family on that table. No individual column is a novelty claim; MOSAIC's contribution is the complete final row.

## 4 Adaptive Universal-Attacker Certificate

For a bounded vector $v$, define the exact confidence-set support function

$$\Phi(\widehat{p}, \epsilon; v) = \max_{p \in \Delta_K : \|p - \widehat{p}\|_1 \leq \epsilon} p^\top v. \tag{4}$$

It is obtained by moving at most $\epsilon/2$ mass from low to high coordinates of $v$, or from an equivalent linear-program dual. For attacker assignment $a \in [G]^L$, let $v_{M,a,s}(c) = \sum_z M(c, z)\mathbf{1}\{a(z) = s\}$. We then define

$$\overline{A}(\widehat{P}, M) = \frac{1}{G} \max_{a \in [G]^L} \sum_s \Phi(\widehat{p}_s, \epsilon_s; v_{M,a,s}). \tag{5}$$

**Theorem 1** (Adaptive exact envelope)**.** *On $\mathcal{E}$, simultaneously for every row-stochastic $M$, including a channel selected as an arbitrary function of the same table, $A(P, M) \leq \overline{A}(\widehat{P}, M)$. For fixed $M$, the right-hand side is the exact supremum over the product of the stated $\ell_1$ confidence sets.*

The proof writes the Bayes attacker as a maximum over deterministic assignments. For a fixed assignment the source confidence sets separate, giving one support function per source; all maxima are finite and commute. The implication is pointwise over the complete stochastic-channel space, which is why adaptive selection needs no channel-count correction.

## 5 Data-Certified Shift Membership

The structured model in Eq. (3) is useful only when membership is evidenced rather than presumed. Let a labeled bridge sample from the target population define simultaneous confidence sets $\mathcal{D}_{s,y}$ for its fine-token laws $q_{s,y}$. Write

$$h_{s,y}(w) = \max_{p \in \mathcal{C}_{s,y}} p^\top w, \qquad \underline{q}_{s,y}(z) = \min_{q \in \mathcal{D}_{s,y}} q(z). \tag{6}$$

For each label, the bridge program chooses retained mass $t$ and $W = tT$ by solving

$$\begin{aligned} \max_{t,W} \quad & t \\ \text{s.t.} \quad & W \geq 0, \quad W\mathbf{1} = t\mathbf{1}, \quad 0 \leq t \leq 1, \\ & h_{s,y}(W_{:z}) \leq \underline{q}_{s,y}(z) \quad \forall s, z. \end{aligned} \tag{7}$$

The support functions and coordinate lower bounds have exact LP duals, so Eq. (7) is one linear program rather than a transform search.

**Theorem 2** (Adaptive finite-sample bridge)**.** *Suppose the reference and bridge confidence sets jointly cover every $p_{s,y}$ and $q_{s,y}$. Any feasible $(\widehat{t}, \widehat{W})$ with $\widehat{t} > 0$, including the same-table maximizer, defines $\widehat{T} = \widehat{W}/\widehat{t}$ and residual laws satisfying*

$$q_{s,y} = \widehat{t}\, p_{s,y} \widehat{T} + (1 - \widehat{t}) r_{s,y} \quad \forall s. \tag{8}$$

*Moreover, Eq.* (7) *returns the largest retained mass certified uniformly over the product confidence region.*

Indeed, feasibility and coverage imply $q_{s,y}(z) \geq p_{s,y}^\top \widehat{W}_{:z}$ coordinatewise. The difference is nonnegative and sums to $1 - \widehat{t}$, so normalization constructs $r_{s,y}$. Conversely, robust membership requires exactly these coordinatewise inequalities; maximizing the reference support and minimizing the bridge coordinate yields Eq. (7). Thus the learned singleton transform $\{\widehat{T}_y\}$ and contamination $\widehat{\eta}_y = 1 - \widehat{t}_y$ can be inserted into the certificates below on the same event. If a required bridge source–label stratum is absent, its confidence set is the full simplex, every coordinate lower bound is zero, and Eq. (7) forces $\widehat{t} = 0$.

## 6 Certification under Pre-Release Shift

We use three names consistently: the *data-certified bridge* tests shift membership, the *transform-exact certificate* enumerates the learned shift geometry, and the *capacity fallback* upper-bounds it when that geometry cannot be enumerated.

Define the arbitrary differential-shift capacity

$$\mathrm{Cap}_G(M) = \frac{1}{G} \max_{a \in [G]^L} \sum_s \max_c v_{M,a,s}(c). \tag{9}$$

This equals $\sup_{r_1,\ldots,r_G} A(R, M)$. For $G = 2$, its normalized value is the Dobrushin coefficient, the largest total variation distance between two rows of $M$.

For a transform polytope with registered extremes $\mathcal{V}_y = \mathrm{ext}(\mathcal{T}_y)$, define the shifted envelope

$$\begin{aligned} \overline{A}_y^X(M) = \frac{1}{G} \max_{T \in \mathcal{V}_y,\, a \in [G]^L} \sum_s \Big[ & \\ & (1 - \eta_y)\Phi(\widehat{p}_{s,y}, \epsilon_{s,y}; T v_{M,a,s}) \\ & + \eta_y \max_c v_{M,a,s}(c) \Big]. \end{aligned} \tag{10}$$

**Theorem 3** (Transform-exact structured-shift certificate)**.** *On $\mathcal{E}$, every same-table selected $M$ and every external law* *in Eq.* (3) *satisfy* $A_y(Q, M) \le \overline{A}_y^X(M)$*. For fixed* $M$*, the right-hand side is the exact supremum over the stated confidence sets, transform polytope, retained masses, and arbitrary residual laws.*

For fixed $T$, retained mass, and attacker assignment, the source confidence sets and residual simplexes separate. Their support values are respectively $\Phi(\widehat{p}, \epsilon; Tv)$ and $\max_c v(c)$. The latter is no smaller, so the worst retained mass is $1 - \eta_y$. The support objective is convex in $T$; its maximum over a polytope occurs at an extreme. These finite maxima commute, proving exactness. Since the statement holds pointwise for every $M$, it survives adaptive channel selection.

When transform extremes cannot be enumerated, a capacity-fallback corollary is available. Define $\rho_T(M) = \max_c \mathrm{TV}((TM)(c, \cdot), M(c, \cdot))$ and $B_y(M) = \min\{\mathrm{Cap}_G(M), \overline{A}_y(\widehat{P}, M) + \rho_{\mathcal{T}_y}(M)\}$. Then

$$A(Q, M) \le (1 - \eta_y) B_y(M) + \eta_y \, \mathrm{Cap}_G(M). \tag{11}$$

The exact envelope in Eq. (10) is never larger. When $\eta_y = 0$ and $TM = M$, both reduce to the reference certificate; when $\eta_y = 1$, both reduce to exact distribution-free capacity.

Because the fallback separately upper-bounds every law in the same structured class, both transform-exact source and utility values are pointwise no larger than their fallback values. "Never worse" is therefore an implementation check implied by the theorem; experiments measure how often tightness changes the release decision. Proofs are in the supplement.

For decoder $g : [L] \to [J]$, let $u_y(c) = \sum_z M(c, z) \mathbf{1}\{g(z) \ne y\}$. The matching exact conditional error certificate is

$$\overline{E}_{s,y}^X(M, g) = \max_{T \in \mathcal{V}_y} \Big[(1 - \eta_y) \, \Phi(\widehat{p}_{s,y}, \epsilon_{s,y}; Tu_y) + \eta_y \max_c u_y(c)\Big]. \tag{12}$$

It is the exact supremum over the same confidence and shift sets. Thus $M$, $g$, within-label source distinguishability, and worst-stratum utility can all be selected on one event.

**Theorem 4** (No free lunch)**.** $\mathrm{Cap}_G(M) = 1/G$ *if and only if all rows of* $M$ *are identical. For two sources, if two fine tokens carry different task labels and incur row-wise decoder errors at most* $e_0$ *and* $e_1$*, then the normalized capacity is at least* $1 - e_0 - e_1$.

Consequently, unrestricted source-specific shift permits exact erasure only by making the public token independent of the fine token. A bounded $\eta_y$ is therefore an identifiable assumption of the deployment contract, not a hidden technical convenience.

There is a separate identifiability boundary for external auditing. Suppose a required source is absent and let $A_{\mathrm{miss}}(q_0, M) = \sup_{q_1 \in \Delta_K} A((q_0, q_1), M)$.

**Theorem 5** (Missing-source audit)**.** *For a release channel* $M$ *fixed independently of the missing-source law and audit randomness, any protocol with false-certification probability at most* $\delta$ *uniformly over the unobserved law* $q_1$ *can claim* $A((q_0, q_1), M) \le \tau < A_{\mathrm{miss}}(q_0, M)$ *with probability at most* $\delta$.

The observed audit distribution is identical for every missing law; choosing a simplex maximizer makes every stronger claim a false certification. The same argument makes the utility bound for a missing source–label stratum at least $\max_c(M\ell_y)(c)$. Missing required strata are therefore unestimable, not evidence of safety; the supplement gives the full proof.

## 7 Global Optimization and Abstention

For fixed decoder, support functions, transformed scores, and residual maxima have linear epigraphs. Enforcing all transform extremes and $G^L$ attacker assignments gives one LP in the $KL$ channel entries; enumerating $J^L$ decoders yields the global finite-alphabet optimum. The capacity fallback uses one mixed-integer LP per decoder. Acceptance requires solver optimality, valid primal residuals, and agreement with separately recomputed certificates.

**Release protocol.** Fix tokenizers, alphabets, thresholds, split roles, and confidence allocation before viewing certification or bridge folds. Construct simultaneous regions, solve Eq. (7), and enumerate decoder LPs subject to every source constraint. Replay each returned certificate, then select the feasible registered tokenizer with minimum certified error under a fixed tie break. Zero retained mass, missing support, no feasible row, or a failed numerical check returns ABSTAIN; otherwise only $(M, g)$ and $Z$ are released.

With $R = \max_y |\mathcal{V}_y|$, the direct formulation is polynomial in $K$ for fixed $(G, J, L, R)$ but exponential in the deliberately small $L$ through attacker and decoder enumeration. Larger public alphabets require separation or column generation. Endpoint enumeration, population identities, dense channel grids, and randomized exact-versus-fallback comparisons check the encoding; the supplement gives complexity and power-curve details.

## 8 Hash-Locked Evaluation

**Synthetic confirmation.** Before outcomes, we SHA-256 locked the protocol, code, evaluators, seed exclusions, gates, and theory curve. Two source–label populations, two shift regimes, eight rules, and up to five sample sizes use 1,000 independent tables per cell and familywise $\delta = .05$. A separate population evaluator grades every selected channel, and replay scripts reconstruct every decision, risk, interval, and optimizer check. The primary contrasts test unsafe plug-in selection and safe retention against held-out certification, LTT, and deterministic release; every registered cell remains reportable.

## 9 Locked Synthetic Results

All preregistered gates pass. Figure 2 reports every implementable rule in the primary cells; "fallback" is MOSAIC's generic capacity certificate. Independent replay recomputes all 64,000 method receipts, 128,000 population risks, decisions, and aggregate intervals with zero mismatch.

The paired safety difference is 42.1 points (95% bootstrap interval $[39.0, 45.2]$), with 421 plug-in-only violations and no reverse discordance. Across 8,000 fallback certification

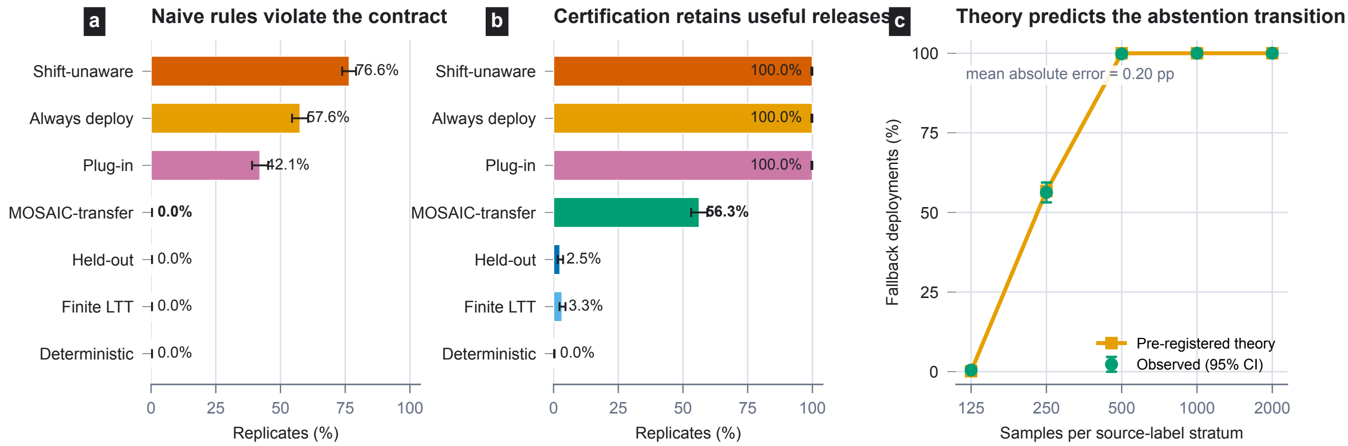


Figure 2: Hash-locked confirmation with 1,000 independent certification tables per cell. Error bars are exact 95% Clopper–Pearson intervals. (a) At the safety boundary, plug-in continuum selection falsely accepts 42.1% of tables; MOSAIC abstains with no false acceptance. Ignoring shift is worse. (b) In the retention cell, the fallback safely deploys on 56.3%, versus 2.5% for held-out certification, 3.3% for finite LTT, and 0% for deterministic MOSAIC. (c) Deployment follows the preregistered theory curve with 0.20 percentage-point mean absolute error. Exact 95% Clopper–Pearson intervals are shown.

tables, no accepted release violates the exact population contract. In the retention cell, fallback safe deployment exceeds held-out, finite-LTT, and deterministic rates by 53.8, 53.0, and 56.3 points. No deterministic channel is feasible. These finite-run facts support, but do not replace, the $1-\delta$ guarantee.

A separately locked 5,000-table refinement evaluates exact shift geometry on the same tables. At $n = 125$, transform-exact certification safely deploys on 53.0% versus 0.3% for fallback; at $n = 250$, the rates are 99.9% and 57.8%. The exact objective is no larger on all 5,000 pairs, as theory requires, and all 10,000 method-cell decisions have zero false acceptances. Full cell tables, threshold sensitivity, amendment history, and audits are in the supplement.

## 10 Guarantee Semantics

The probability $1-\delta$ is over simultaneous reference and bridge regions. On that event, the conclusion is uniform over optimized channels and every law in the certified shift class; it does not assume that a validation interval covers an external metric. A frozen tokenizer's table covers the channel continuum and every decoder without a channel-count correction. Separately learned tokenizers or erasers have different tables and therefore receive a registered familywise allocation. MOSAIC pays for statistical objects, not every point optimized inside one covered object.

**Threat surface.** The attacker knows $M$ and may apply any rule to $Z$. The source-blind software channel is the only public representation interface and samples once per immutable item; persistent repeats reveal no more than one $Z$. If $r$ fresh draws are permitted, certification must use product channel $M^{\otimes r}$; unbounded draws can reveal its row. Transcripts over distinct items require a joint-law certificate. Releasing $C$, original features, source-dependent metadata, mutable identifiers, or another side channel is outside scope.

## 11 Limitations and Scope

The bridge requires fixed-size i.i.d. or exchangeable reference and target data for every required source–label stratum and does not cover later drift. Tokenizers are fixed independently of certification; large alphabets may need structured regions or column generation. MOSAIC trades reusable embeddings for a certifiable task interface and durable persistence state. It certifies within-label source inference from $Z$, not label-prevalence leakage, separate interfaces, every downstream harm, or utility forbidden by the contract.

## 12 Conclusion

MOSAIC treats editing as release-mechanism design. A pre-channel confidence event permits same-table continuum optimization; the bridge separates common drift from differential contamination, and impossibility results identify when abstention is unavoidable. Locked experiments show both outcomes: refusal at an unsafe boundary and much higher safe retention when target shift geometry is certifiable.

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